Jorge Benavides Macías

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Work Experience

System Embedded Engineer — Sept. 2025 – Present
Clue Technologies — *Malaga, Spain*

- Drive systems validation and hardware stress testing for general-purpose mission computers in avionics and defense.
- Developed customized C and Python stress-testing software to validate aerospace communication protocols, including RS485, RS422, ARINC 818, Ethernet and fiber optics.
- Engineered video processing benchmarks and hardware stress tools to evaluate maximum thermal limits, power consumption, and peak system reliability.
- Validated low-level software stacks including Bare-metal applications, VxWorks RTOS environments, and U-Boot bootloaders.

Junior Cybersecurity & Microelectronics Engineer — Jul. 2024 – Sept. 2025
InnovaIRV — *Malaga, Spain*

- Led cybersecurity needs assessments for over 15 Andalusian technology companies through executive/CISO interviews, identifying critical gaps in network security, access control, and incident response.
- Authored a 150-page technical report with prioritized countermeasures as a regional cybersecurity roadmap for Andalusia's Government.
- Technical execution of a full project lifecycle (€600k budget) for *Impulso&Crece2024* digitalization initiative funded by regional authorities.

R&D Jr. Electronic Engineer — Mar. 2024 – Jul. 2024
Quandum AeroSpace S.L. — *Malaga, Spain*

- Designed a 15W wireless charging system for vertical wind turbines using Altium, reducing component count by 30% and optimizing overall PCB layout size.
- Implemented RTOS-based motor control on ESP32.
- Collaborated on mechanical design (CATIA), PLC automation, and Linux-based version control system administration.

Projects

PCB Design for Robotic Gripper | *C, ARM, PSoC, LabView, Eagle* | DOI

- Designed and manufacturing a PCB for data acquisition from resistive tactile sensors integrated into a robotic manipulator's gripper. This involved full process design, including component selection, material procurement, and PCB trace routing, it incorporated a PSoC in a QFN package.
- Developed a master-slave architecture in C for ARM, collecting data from multiple sensors via SPI and UART communication, with debugging facilitated by LabView.

LaCaja | *C, MQTT, MongoDB, JavaScript, HTML, CSS, NodeRED* | News – Web – Repository

- Earned the opportunity to visit Google's Malaga offices and present the project.
- Crafted an AI-driven escape room experience featuring "Chiquito de la Calzada", offering participants a chance to win a position at Google.
- Designed 2/5 games (software & hardware), managing PCB soldering and sensor connections.

Education

University of Malaga — Malaga, Spain
Master's Degree in electronic systems for smart environments — *Sep. 2024 – Present*

University of Malaga — Malaga, Spain
Bachelor in electronics, robotics & mechatronics. Major in robotics & automation — *Sept. 2018 – Sept. 2023*

Technical Skills

Languages: Spanish (native), English proficient (B1)
Embedded Systems: STM32, ESP32, AVR, RTOS, PCB Design (Altium/Eagle/KiCAD) and 3D CAD Model (CATIA, SolidWorks)
Programming: C/C++, Python, Bash, VHDL
Tools: Git, Docker, Vivado, LabVIEW, MATLAB, Emacs